

Formalizing Pre-Event Staging Decisions in Emergency Management Practice with Implications for AI Decision Support

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Abstract

AI systems for disaster logistics have proliferated, yet most research focuses on algorithmic innovation while leaving problem specification underexamined. This inverts a foundational principle from sequential decision theory: *model [the problem] first, then solve*. We address this gap by systematically extracting formal decision elements from semi-structured interviews with 32 U.S.-based emergency managers, focusing on pre-event staging—the positioning of resources before anticipated disaster impact. Adapting Powell’s (2022) universal framework for sequential decision problems, we identify six elements that characterize staging decisions: objectives, constraints, information structure, temporal structure, action space, and evaluation criteria. Our analysis reveals decision structures that standard optimization formulations miss: threshold-constrained hierarchical objectives where life safety must be satisfied before secondary objectives are optimized; dual spatial constraints creating annular feasible regions for staging locations; threshold-based forecast utilization rather than continuous probability integration; and asymmetric costs favoring over-preparation. We translate these findings into four actionable design implications for AI systems supporting emergency management, demonstrating a replicable methodology for grounding computational approaches in practitioner expertise.

1 Introduction

AI applications for humanitarian assistance, emergency management, and disaster relief have expanded rapidly, spanning demand forecasting, resource pre-positioning, evacuation routing, and damage assessment (McEntire et al. 2024). Yet a persistent gap separates these technical contributions from operational deployment (Shariati, L’Hermitte, and Trent 2025). We argue that this gap stems not from algorithmic limitations but from inadequate problem specification: researchers optimize objectives, satisfy constraints, and create information structures that diverge from how Emergency Managers (EMs) actually make decisions.

This pattern inverts a foundational principle from sequential decision theory: “model first, then solve” (Powell 2022). Accurate problem specification must precede algorithm design; disaster logistics research has largely proceeded in reverse—sophisticated solution methods applied to under-specified or misspecified problem formulations (Pedraza-Martinez and Van Wassenhove 2016; Shariati, L’Hermitte,

and Trent 2025). The result is systems that may optimize the wrong objectives, ignore critical constraints, or assume information availability that practitioners lack.

We focus on *pre-event staging*: decisions about where and when to position personnel, equipment, and supplies in anticipation of disaster impact. Staging represents a consequential yet bounded decision context well-suited for formal analysis: clear temporal structure (forecasts trigger staging windows), observable information sources (weather predictions, resource inventories), tractable action space (locations, quantities, timing), and direct downstream consequences. Pre-event staging exemplifies sequential decision-making under uncertainty—decision-makers commit resources based on probabilistic forecasts then adapt as information evolves—making it an ideal case for demonstrating how practitioner insights can inform formal problem specification.

This paper makes three contributions. First, we present a **systematic extraction of decision elements** from semi-structured interviews with 32 U.S.-based emergency managers. Adapting Powell’s (2022) five universal elements for sequential decision problems, we develop a six-element framework suited for qualitative extraction: objectives, constraints, information structure, temporal structure, action space, and evaluation criteria.

Second, we **identify decision structures that standard formulations miss**. Our analysis reveals that practitioners employ threshold-constrained hierarchical objectives where life safety must be satisfied before secondary objectives are optimized—distinct from both weighted trade-offs and strict lexicographic maximization; face dual spatial constraints creating annular feasible regions; consume forecasts through discrete action triggers rather than continuous probability distributions; and operate under asymmetric loss functions where under-preparation carries higher costs than over-preparation.

Third, we **translate these findings into design implications** for AI systems supporting emergency management, mapping practitioner-grounded findings to formal constructs and identifying gaps between current computational approaches and operational requirements.

Our goal is not to propose new algorithms but to demonstrate a methodology for the “model first” phase that deserves significantly more attention from AI research. We of-

fer pre-event staging as an exemplar, with the approach generalizable to other emergency management decision types. Section 2 reviews related work; Section 3 describes our methods; Section 4 presents findings; Section 5 articulates design implications; and Section 6 discusses limitations and future directions.

2 Related Work

AI/ML for Disaster Logistics and Pre-Positioning

AI applications for disaster relief logistics have matured substantially, with two-stage stochastic programming emerging as the dominant paradigm for pre-positioning problems (Rawls and Turnquist 2010; Salmerón and Apte 2010; Kamyabniya et al. 2024). Recent extensions incorporate distributionally robust optimization under demand ambiguity (Bozorgi-Amiri, Jabalameli, and Mirzapour Al-e Hashem 2013; Yang et al. 2021) and multi-period dynamics (Ni, Shu, and Song 2018). While computationally sophisticated, these approaches assume objectives, constraints, and information structures are known *a priori*—the gap Powell (Powell 2022) identifies as the distinction between modeling and solving.

Shariati, L’Hermitte, and Trent (2025) provide compelling evidence of this gap in a systematic review of 97 prepositioning studies. Their findings are stark: cost minimization dominates as the objective (77 of 97 papers), yet shortage costs—proxies for human suffering—are defined through “arbitrary” penalty multipliers raising “ethical concerns about assigning economic value to individuals’ suffering” (p. 8). Multi-objective formulations appear in only 25 papers. Most critically, approximately 90% of studies developed models without reported practitioner interaction, and only two achieved real-world implementation.

This practitioner gap manifests in specific modeling choices that our findings will contest. The literature predominantly assumes: (1) weighted-sum objectives where life safety can be traded against cost; (2) facility location constraints focused solely on proximity rather than safety buffers; (3) continuous probability distributions as information representation; and (4) symmetric loss functions treating over- and under-preparation equally (Shariati, L’Hermitte, and Trent 2025). Each assumption represents a testable hypothesis about practitioner decision-making that remains largely untested.

Formal Frameworks for Sequential Decisions

Powell (Powell 2022) provides a universal framework for sequential decision problems comprising five elements: state variables (physical R_t , informational I_t , belief B_t), decision variables and constraints, exogenous information W_{t+1} , transition function, and objective function. For problems with competing objectives, lexicographic MDPs offer a principled alternative to weighted-sum formulations when strict priority orderings exist (Wray, Zilberstein, and Mouaddib 2015). The disaster logistics literature has engaged these frameworks primarily at the solution stage; limited work has applied them in reverse—extracting and validating problem

structure from practitioner accounts before proceeding to algorithms.

Emergency Management as Under-Studied AI Context

Crisis informatics has generated rich understanding of how citizens and digital volunteers use technology during disasters (Palen and Anderson 2016), yet professional EMs’ perspectives remain underexamined. Studies engaging practitioners reveal pragmatic orientations toward “good enough” solutions given chronic resource constraints (Tapia and Moore 2014). These constraints are structural: 57% of local emergency management agencies operate with one or fewer full-time employees, with 56% of directors holding additional duties (Argonne National Laboratory and National Preparedness Analytics Center 2025). Such realities shape how practitioners conceptualize decisions—and what AI systems might feasibly support. The pattern of using practitioners as data sources while excluding them from problem specification may explain why sophisticated optimization models fail to achieve operational deployment.

We address this gap by providing empirically grounded problem specifications derived from practitioner accounts. Our contribution bridges crisis informatics’ contextual richness with decision science’s formal rigor, demonstrating systematic translation from qualitative interview data to Powell’s universal framework. Where Shariati, L’Hermitte, and Trent (2025) document how the literature models prepositioning decisions, we document how practitioners actually make them—addressing the qualitative foundation their review identifies as missing.

3 Methods

We conducted semi-structured interviews with 32 U.S.-based EMs to elicit detailed accounts of critical decision-making during disaster events. Participants were recruited using purposive sampling with snowball recruitment (Patton 2015), achieving geographic diversity across 13 states (AZ, CA, DC, FL, IL, LA, MA, MI, NC, NV, NY, PA, TX) and role diversity spanning county EMs, state-level coordinators, and hospital emergency administrators. Appendix Table 2 summarizes key participant information. Years of experience ranged from 5 to 45 ($M=20.7$, $Mdn=21.5$). Interviews (60–90 minutes, conducted via Zoom) employed the critical incident technique (Flanagan 1954): rather than posing abstract questions, we asked participants to walk through specific high-stakes decisions, probing for decision processes, information sources, constraints, and outcomes. All materials received IRB approval.

Analytic Scope: Pre-Event Staging

The interview script elicited multiple critical decision types, including resource allocation, evacuation timing, and protective actions and (Appendix B). This paper focuses on *pre-event staging decisions*—anticipatory positioning of resources and personnel before a forecasted disaster materializes. We do so for three reasons: (1) coherent temporal structure with clear decision triggers, (2) bounded action

Table 1: Pre-Staging Analytic Sample (n=11 participants, 33 quotations)

ID	Jurisdiction	Exp.	Disaster text	Con-
<i>Florida—Hurricane/Storm Surge (n=4)</i>				
FL2	State/Local	22 yrs	Hurricane; shelter, staging	Ian; UAS
FL3	Local (Other)	15 yrs	Hurricane; EOC activation	EOC
FL4	Local (Other)	18 yrs	Hurricane; hospital resources	hos-
FL6	Local (Municipal)	35 yrs	Hurricane; split-force staging	split-
<i>Pennsylvania—River Flooding (n=3)</i>				
PA2	Local (County)	26 yrs	River flooding; water rescue	
PA3	Local (County)	6 yrs	Water emergency; logistics staging	
PA7	Local (Other)	26 yrs	River flooding; regional coord.	
<i>Other Contexts (n=4)</i>				
DC1	State	12 yrs	Flood; sensor-based triggers	
IL1	Local (County)	22 yrs	Multi-hazard; staging areas	
NC1	Local (Municipal)	12 yrs	T.S. Helene; EOC scoring	
TX1	Contractor	45 yrs	Hurricane; EOC exercises	

Note: Participants represent county/municipal EM (7), state-level coordination (2), hospital EM (1), and contractor/consultant roles (1). Experience: M=20.1 years, range 6–45. Geographic contexts span coastal hurricane zones (FL), inland river systems (PA), and mixed urban/regional jurisdictions (DC, IL, NC, TX).

space amenable to formal specification, and (3) direct relevance to established frameworks in disaster logistics (Pu et al. 2025). Of the 32 participants, 11 provided substantive accounts of pre-event staging, yielding 33 quotations about forecastable disaster events (hurricanes, flooding, wildfires). Table 1 summarizes the subsample we analyzed.

Analytic Framework

Powell identifies five universal elements for modeling any sequential decision problem: (1) state variables, (2) decision variables with constraints, (3) exogenous information, (4) transition function, and (5) objective function (Powell 2022). We adapted this framework for qualitative extraction from practitioner accounts, yielding six elements:

1. **Objectives:** What is the decision-maker trying to achieve? (*maps to objective function*)
2. **Constraints:** What boundaries limit feasible decisions? (*maps to decision variable constraints; separated because practitioners discuss “what limits us” distinctly from “what we can do”*)

3. **Information Structure:** What is known, uncertain, and when does information arrive? (*combines state variables and exogenous information; practitioners do not cleanly separate current knowledge from arriving information*)
4. **Temporal Structure:** When must decisions be made relative to outcomes? How do staging decisions constrain deployment? (*maps to transition function*)
5. **Action Space:** What options are available? (*maps to decision variables; separated from constraints for analytic clarity*)
6. **Evaluation Criteria:** How is decision quality assessed? (*maps to objective function; separated because practitioners articulate goals vs. assessment metrics distinctly*)

This adaptation preserves Powell’s formal structure while accommodating how practitioners naturally describe their decision-making in interviews.

Coding Approach

We employed deductive coding using the six-element framework. Refer to Table 3 for a detailed summary of participant coverage for each decision element in our corpus. For each quotation, two analysts independently coded each element at three levels: *explicit* (practitioner stated directly), *implicit* (inferable from context), or *absent* (not addressed), documenting supporting evidence for each judgment. Disagreements were reconciled through discussion until consensus. We then aggregated within-element patterns across quotations, tracking participant coverage to calibrate confidence in findings. Finally, we mapped observed structures to formal problem classes, identifying alignments and gaps with standard formulations.

4 Findings

Our analysis reveals six patterns that inform formal problem specification. We organize findings by our six-element framework, tagging each to Powell’s elements (Powell 2022). Table 3 (Appendix A) reports participant coverage; elements with $\geq 75\%$ coverage are characterized with higher confidence.

Objectives: Threshold-Constrained Hierarchical Priorities

Practitioners articulated strict priority orderings among objectives rather than weighted trade-offs. One Florida EM stated: “Preservation of life, then preservation of property, then control of the situation” (FL4). Another confirmed: “Life safety is what we do. If the priority is life safety for that operational period, we won’t entertain any requests that aren’t immediate life safety” (PA3).

Critically, participants described *satisfying* life-safety requirements rather than *maximizing* life safety. IL1 framed it as meeting needs: “First and foremost, life safety. What do we have to do for life safety?” FL3 described a binary threshold check: “determining whether there’s an ongoing loss of life or great bodily harm... that is the criteria.” This pattern—checking whether needs are *met* rather than *maximized*—was consistent across participants.

This structure differs from both weighted-sum formulations ($\sum_i w_i f_i(x)$), which allow trade-offs practitioners rejected, and strict lexicographic optimization, which requires *maximizing* life safety before considering any other objective. Instead, practitioners ensure life-safety needs are adequately addressed, then attend to secondary objectives. Formally:

$$\pi^* = \arg \max_{\pi} f_k(\pi) \quad \text{s.t.} \quad f_i(\pi) \geq \theta_i \quad \forall i \in \{1, \dots, k-1\}$$

where objectives are priority-ordered ($f_1 \succ f_2 \succ \dots \succ f_k$) and each θ_i represents a context-dependent satisficing threshold—“enough” rather than “maximum.”

Constraints: Dual-Constraint Spatial Optimization

Unlike facility location problems that optimize for proximity, staging decisions must satisfy two opposing spatial constraints. First, staging locations must be close enough to enable rapid deployment: “We try to have our resources available in each of those areas so that we can get into those areas faster than coming from one centralized location” (FL6). Second, locations must be far enough that staged resources are not themselves impacted: “I had to find locations that were well enough inland, but close enough that I could get those resources [to] a staging area... make sure those first responders that are going to those areas had a safe place to domicile before the storm” (FL6).

These constraints create an *annular* feasible region around the forecast impact zone. Let $d(x, z_t)$ denote distance from staging location x to forecast impact zone z_t :

$$\begin{aligned} d(x, z_t) &\leq d_{\text{deploy}}^{\max} && \text{(deployment time)} \\ d(x, z_t) &\geq d_{\text{safe}}^{\min} && \text{(safety buffer)} \end{aligned}$$

This geometry differs fundamentally from p-median or covering problems, where only proximity constrains solutions.

Information Structure: Threshold-Based Forecast Interpretation

Participants described consuming forecasts through discrete action triggers rather than continuous distributions. PA2 described their trigger structure: “When the [weather service] puts out a warning, that’s when we start making calls... we’re not sitting there watching the percentages tick up.” NC1 developed a scoring system: “I have a spreadsheet... level of threat to life and property, time to meet objectives, impacts to community, geographic area involved, availability of resources... each of those answers are weighted with a score... based upon what that score is, that activates our emergency operations plan and our EOC activation level.” DC1 envisioned AI support: “[The AI] triggers a flood alert when the river gets 3 feet of inundation... ‘Here’s a recommended alert...[do] you want to send it out, yes or no?’”

Rather than maintaining continuous probability distributions, practitioners discretize the information space into categorical states (Watch, Warning, Emergency) that map directly to action triggers:

$$\tau(W_{t+1}) = \begin{cases} \text{Stage Level 1} & \text{if } W_{t+1} \in \mathcal{W}_1 \\ \text{Stage Level 2} & \text{if } W_{t+1} \in \mathcal{W}_2 \\ \vdots & \end{cases}$$

This does not preclude AI systems from processing continuous distributions internally; the NC1 scoring model shows practitioners can work with weighted inputs producing continuous scores—but the *output* triggering action is categorical.

Temporal Structure: Asymmetric Reversibility

A consistent theme was asymmetry between over- and under-preparation. FL4 summarized: “It’s always easier to scale down than scale up.” FL2 explained: “I’m going to overestimate a little bit, because I don’t want to stand up a shelter and say... we’re assuming we’re going to get 500 people and then 1000 require sheltering.” This reflects a fundamental property: resources staged in advance can be demobilized, but resources not staged cannot be materialized on demand.

Qualifier: This asymmetry holds when resources are *locally controlled*. When resources come from shared regional pools, over-staging in one jurisdiction can create under-staging elsewhere. PA7, managing regional water rescue coordination: “One of my responsibilities has been during weather events, to track what teams are engaged, what teams are available, so we can start pre-positioning.” PA3 described hierarchical escalation: “Requests come to us from the local level... a lot of the times at the county level, we simply don’t have the resources to help them. So we go to what’s called a task force.”

For locally-held resources, let x_t denote the staging decision and W_{t+1} the realized severity:

$$L(x_t, W_{t+1}) = \begin{cases} c_{\text{cover}} \cdot (x_t - W_{t+1}) & \text{if } x_t > W_{t+1} \\ c_{\text{under}} \cdot (W_{t+1} - x_t) & \text{if } x_t < W_{t+1} \end{cases}$$

where $c_{\text{under}} \gg c_{\text{cover}}$ for locally-controlled resources. When resources are drawn from shared pools, the asymmetry applies to *aggregate system resources*, not individual allocations.

Additionally, staging decisions constrain subsequent deployment. Resources staged in the wrong location cannot simply be redirected—they represent committed capacity that may delay response if forecasts were incorrect.

Action Space: Authority-Bounded Options

EMs describe their action space as constrained by jurisdictional authority, not just physical resources. PA2 illustrated: “It’s always going to be a voluntary evacuation... just going to be a recommendation”—mandatory evacuation is simply not in this practitioner’s feasible action set. FL2 explained the institutional context: “Everybody in emergency management ultimately works for an elected official of some sort—county commissioner, the mayor, governor.” TX1 noted the advisory role: “I’ve never been the one who had to make the decision, but I’ve been the one who’s had to provide the information to the decision maker, generally an elected official.”

The action space $\mathcal{X}_t$ is defined by authorized authority, not physical possibility. Mutual aid agreements expand the action space but with coordination costs—additional decision variables (requests), constraints (approval processes, lead

times), and costs (administrative burden, reciprocity). This implies that $\mathcal{X}_t$ in any formal model must be *socially constructed*—determined by institutional position, not merely physical inventory.

Evaluation Criteria: Process vs. Outcome Tension

How staging decisions are evaluated revealed tension between process-based and outcome-based criteria, often from different stakeholders. IL1 captured this landscape: “Four things affect every decision: liability, media, funding, politics.” A staging decision that minimizes response time (operationally optimal) might maximize political exposure (politically suboptimal) if it requires visible pre-commitment that proves unnecessary. TX1 illustrated this vividly regarding a COVID medical shelter: “You can get it and not need it, and you’ll be ripped apart in the media and by the general public, or you can not get it and need it, and you’ll be ripped apart by the media and the general public. The only way you could ‘win’ is if you get it and we need it, and nobody wanted that either.”

This suggests the objective function must accommodate multi-stakeholder evaluation. The reward function $R(S_t, x_t)$ may need replacement by a vector-valued function $\mathbf{R}(S_t, x_t) = (R_{\text{operational}}, R_{\text{legal}}, R_{\text{political}}, \dots)$ with potentially non-aligned components.

For modeling, this suggests that the objective function must accommodate multi-stakeholder evaluation rather than optimizing a single operational metric. The reward function $R(S_t, x_t)$ implicitly assumed by most optimization formulations may need to be replaced by a vector-valued function $\mathbf{R}(S_t, x_t) = (R_{\text{operational}}, R_{\text{legal}}, R_{\text{political}}, \dots)$ with potentially non-aligned components.

5 Design Implications for AI Systems

We translate our findings into four design implications for AI staging support. Each maps a finding to formal constructs, identifies gaps with current approaches, and offers recommendations.

Implication 1: Threshold-Constrained Hierarchical Objectives

Finding: EMs articulate strict priority orderings where life safety operates as a threshold constraint—they *satisfy* life-safety requirements before attending to secondary objectives, rather than *maximizing* life safety absolutely (§4.1). This threshold-then-optimize pattern appeared consistently across all 32 participants.

Formal Translation: We distinguish two formulations: (1) *Strict lexicographic optimization* $\pi^* = \text{lex-opt}(f_1, \dots, f_k)$, which maximizes f_1 first, then f_2 among optimal- f_1 policies, etc.; and (2) *Threshold-constrained optimization*, which maximizes f_2 subject to $f_1 \geq \theta_1$, where θ_1 may be less than $\max_{\pi} f_1(\pi)$. Our interviews indicate EMs operate according to the threshold-constrained formulation (Eq. 1 in §4.1).

Gap: Most disaster logistics formulations use weighted-sum objectives $\max \sum_i w_i f_i(\pi)$, allowing trade-offs between life safety and cost. EMs explicitly reject such trade-

offs. Strict lexicographic formulations capture priority ordering but may over-constrain by requiring absolute maximization of higher-priority objectives.

Recommendation: AI systems should enforce life-safety thresholds as hard constraints before optimizing secondary objectives. Thresholds should be elicited from EMs and may vary by incident type, operational phase, and available resources. Interfaces should clearly communicate when thresholds are met versus unsatisfied.

Implication 2: Dual-Constraint Spatial Modeling

Finding: Staging locations must satisfy simultaneous proximity and safety constraints, creating an annular feasible region (§4.2).

Formal Translation: The constraint set includes both upper and lower bounds on distance from the forecast impact zone (Eq. 2 in §4.2).

Gap: Facility location literature typically models only proximity constraints (coverage radius, distance minimization). The safety buffer preventing staging locations from becoming disaster victims is rarely modeled explicitly.

Recommendation: Staging optimization must model the safety buffer explicitly. GIS tools should visualize both the outer boundary (deployment reach) and inner boundary (safety zone). Feasible staging areas are the annular region between these boundaries.

Implication 3: Threshold-Based Forecast Interfaces

Finding: EMs consume forecasts through discrete action triggers rather than continuous probabilities (§4.3). Practitioners described scoring systems and activation levels tied to categorical states, not probability values.

Formal Translation: We distinguish the *computational layer* and *interface layer* of AI forecast systems. Internally, the AI may process continuous distributions $P(W_t | \mathcal{I}_{t-1})$ for optimal decision-theoretic computation. However, output to human decision-makers should be transformed through a threshold function τ that maps distributions to categorical triggers:

$$B_t = \tau(P(W_t | \mathcal{I}_{t-1})) \in \{\text{Monitor, Stage-1, Stage-2, Full-Activation}\}$$

The function τ should be calibrated to agency-specific protocols, not generic probability cutoffs. Optimal computation and effective human communication require different representations.

Gap: AI systems typically output continuous predictions (distributions, confidence intervals). EMs need discrete, actionable triggers that map directly to their decision protocols.

Recommendation: Forecast systems should surface categorical staging triggers (e.g., “Stage 2 Recommended”) alongside or instead of raw probabilistic outputs. The AI may leverage full probabilistic forecasts internally, but human-facing interfaces should present threshold-based categorical recommendations developed collaboratively with agencies.

Implication 4: Asymmetric Loss Functions

Finding: For locally-controlled resources, over-staging carries substantially lower costs than under-staging (§4.4). However, when resources come from shared pools (mutual aid, task forces), over-staging in one jurisdiction can create under-staging elsewhere.

Formal Translation: The expected loss function is asymmetric:

$$\mathbb{E}[L] = c_{\text{cover}} \cdot \mathbb{P}(\text{over}) \cdot E[\text{excess}] + c_{\text{under}} \cdot \mathbb{P}(\text{under}) \cdot E[\text{shortfall}]$$

where $c_{\text{under}} \gg c_{\text{cover}}$ for locally-controlled resources. When resources are drawn from a shared pool across jurisdictions $\mathcal{J}$:

$$\sum_{i \in \mathcal{J}} x_{it} \leq R_{\text{pool}}$$

the asymmetry applies to *aggregate system resources*, not individual allocations made in isolation.

Gap: Symmetric loss assumptions (e.g., squared error) are common, treating over- and under-preparation as equally costly. Most formulations also treat jurisdictional decisions independently, ignoring coupling from shared resource pools.

Recommendation: Explicitly model asymmetric costs. When uncertainty is high and resources are local, systems should recommend over-preparation. When resources are shared, optimization should coordinate across jurisdictions so asymmetric preferences apply at the system level. Interfaces should make asymmetry visible (“Staging 20% above forecast has cost X; 20% below has cost Y”) and indicate resource pool scope (local vs. shared).

6 Discussion

We provide three contributions to AI research for humanitarian assistance and disaster relief. First, we demonstrate a systematic methodology for translating qualitative practitioner accounts into formal decision elements via Powell’s (2022) universal framework. We address a persistent literature gap: while sophisticated optimization methods abound, their problem specifications often lack empirical grounding in practitioners’ reality.

Second, we identify four structural features of pre-event staging decisions that standard formulations miss: threshold-constrained hierarchical objectives where life safety must be satisfied before secondary objectives are optimized, dual spatial constraints creating annular feasible regions, threshold-based forecast utilization, and asymmetric loss functions favoring over-preparation. Each represents a divergence between existing assumptions in the literature and practitioner values and needs.

Third, we translate these findings into actionable design implications. By rigorously specifying stakeholder problems and needs before designing algorithms, we increase the likelihood that AI systems will actually support, rather than complicate, emergency management practice.

Limitations

Our findings should be interpreted with several limitations in mind. First, our analytic corpus of 33 quotations from

11 participants addressing pre-event staging, while sufficient for pattern identification, may not capture the full diversity of staging practices across the U.S. emergency management landscape. Second, our interviews were designed to elicit rich, open-ended accounts of critical decisions rather than to systematically probe formal decision elements. We made this choice to capture how practitioners naturally describe their decision-making, avoiding the risk that structured elicitation would impose researcher categories. The trade-off is that our six-element framework was applied retrospectively, and some elements—particularly Evaluation Criteria (25% coverage)—may be under-represented.

Third, and most significantly, emergency management practitioners have not validated our formal translations. We mapped qualitative accounts to formal constructs through our interpretation; emergency managers might disagree with our translations or find them incomplete. Validation with practitioners is essential future work.

Finally, pre-event staging represents only one decision type among many that emergency managers face. Our findings may not generalize to decisions with different temporal structures (e.g., EOC activation, real-time resource dispatch) or information environments (e.g., no-notice events). We plan to address a broader scope of EM decisions in forthcoming work.

Future Work

We envision three extensions of this research. First, we plan to apply our methodology to additional decision types—evacuation timing, EOC activation, mutual aid requests—to assess generalizability and identify decision-type-specific structures. Second, we will conduct validation studies with EMs, presenting our formal translations and soliciting feedback on accuracy and completeness. Finally, we aim to prototype AI systems implementing our recommended specifications and evaluate them in simulated and tabletop exercise contexts, with the ultimate goal of creating a deployment-ready systems.

7 Conclusion

AI systems for disaster response require accurate problem specification before algorithmic innovation can be productive. The maxim “model first, then solve” (Powell 2022) is a principle that disaster logistics research has too often inverted. We provide a methodology for the “model first” phase, demonstrating how qualitative practitioner accounts can inform rigorous formal problem specifications on the exemplar of pre-disaster event resource positioning. We hope this work encourages closer collaboration between AI researchers and emergency management practitioners. The sophistication of algorithms means little if they optimize the wrong objectives, ignore binding constraints, or assume information structures that practitioners lack. By grounding problem specifications in practitioner expertise, we can build AI systems that support human judgment in the critical decisions that protect communities from disasters.

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A Extended Methods

All Participant Demographics

Table 2: Participant Data

ID	Type of EM	Jurisdiction Scope	Location	Years	AI Fam.	AI Usage Frequency	
						General	EM work
Research	LA1	N/A (Researcher)	LA	20	A little	Rarely	Rarely
	WW1	National	VA	21	A lot	Often	Rarely
	AZ1	N/A (Researcher)	AZ	5	A little	Often	Often
Research & Leadership	FL2	State, Local (Other)	FL	22	A lot	Rarely	Rarely
	MI1	Local (County)	MI	16	A lot	Often	Often
Leadership	PA4	Local (County)	PA	33	A lot	Very freq	Very freq
	DC1	State	DC	12	A lot	Often	Never
	NC1	Local (Municipal)	NC	12	A little	Rarely	Rarely
	PA2	Local (County)	PA	26	A lot	Rarely	Rarely
	FL3	Local (Other)	FL	15	A lot	Very freq	Very freq
	IL1	Local (County)	IL	22	A little	Often	Often
	PA8	Local (Other)	PA	25+	A little	Often	Rarely
	PA1	Local (County)	PA	13	A lot	Often	Rarely
	TX1	Contractor	TX	45	A lot	Often	Often
	MA2	State	MA	19	A lot	Often	Often
EM Partner & Leadership	PA7	Local (Other)	WV, PA	26	A little	Rarely	Never
	PA10	Local (Municipal)	PA	38	A little	Rarely	Rarely
	PA03	Local (County)	PA	6	A little	Rarely	Rarely
	NW4	National	IL	25	A lot	Often	Often
EM Partner	PA6	Local (County)	PA	26	A lot	Often	Often
	FL6	Local (Municipal)	FL	35	A little	Often	Rarely
Practitioner	PA5	Local (Other)	PA	12	A lot	Very freq	Very freq
	WW2	Worldwide	ID	25	A little	Rarely	Never
	FL4	Local (Other)	FL	18	A little	Rarely	Rarely
	NY1	Local (Municipal)	NY	17	A lot	Often	Rarely
	CA1	Contractor	CA	30+	A little	Never	Never
	NW2	Worldwide	ME	25	A little	Often	Often
	NW3	Contractor	FL	12	A lot	Never	Never
	NW5	Contractor	NY	25	A lot	Very freq	Often
	DC2	Retired	PA	23	A lot	Very freq	Often
Practitioner & Research	FL1	State, Local (Other)	FL	9	A lot	Very freq	Never
	FL8	State, Local (Other)	FL	5	A lot	Often	Rarely

Pre-staging Analysis Data

Analytic Sample Characteristics Our analytic corpus comprises 33 quotations from 11 participants, visualized in Table 1. Geographic contexts include Pennsylvania river flooding (4 participants), Florida hurricane/storm surge (3), Massachusetts storms and planned events (1), New York City hurricanes (1), Arizona multi-state deployment (1), and federal/international operations (1). Participants represented diverse organizational contexts: county and municipal emergency management (5), hospital/healthcare systems (1), state transportation/transit agencies (1), university-based disaster response teams (1), and federal/multi-jurisdictional operations (3).

Six-Element Extraction Protocol

For each quotation, the first author applied the six-element framework, coding each element according to the following criteria:

Explicit: The participant directly states the element. Example (Objective Function): “[We want to] get into those areas faster than coming from one centralized location” (FL6)—response speed is explicitly stated as the goal.

Decision Element	Participants	Coverage
Information Structure	10/11	91%
Constraints	8/11	73%
Temporal Structure	9/11	82%
Action Space	9/11	82%
Objective Function	8/11	73%
Evaluation Criteria	3/11	27%

Table 3: Participant coverage by decision element. Elements with coverage $\geq 75\%$ are characterized with higher confidence.

Implicit: The element is inferable from context but not directly stated. Example (Objective Function): A participant describes staging resources “so that we can do rescues” without stating that rescue capability is the objective—the goal is implicit in the described action.

Absent: The quotation provides no information about the element, either stated or inferable.

For all non-absent codings, we documented the specific text supporting the coding. This documentation enabled cross-quotation synthesis and provides an audit trail for our findings.

B Interview Script

Background

Let’s start with your work background.

1. In a few sentences, can you please describe your work experience related to emergency management?
2. Roughly how many disaster response efforts have you participated in?
3. What types of disaster events have you responded to?
4. How would you describe your current position and job responsibilities?
5. *If not answered in 4* What is your typical role in disaster response efforts?
6. *If indicated in their form response that they use AI in their job* Can you please describe how you use AI in your job?

Decision-making Scenarios

Now that I have a better understanding of your job functions, I’d love to learn more about specific decisions you make in the context of your job.

7. Can you please describe what you consider to be a critical decision in emergency management that you have been a part of?
 - a. In that scenario, can you please tell me about the decision process?
 - b. What was your role in the process?
 - c. What data or factors did you consider when making your decision?
 - d. What information were you given during that scenario?
 - e. What did you know about the situation before you were presented with this decision scenario?
 - f. What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - g. Did you decide in a team or on your own?
 - h. Once you made a decision, can you run me through what happens next?
 - i. Who or what was responsible for the execution of the next steps?

Decision-making Scenarios with Resource Constraints

8. Have you ever encountered a situation where the number of requests for a resource exceeded the resources available?
 - a. In that scenario, can you please tell me about your decision process for allocating resources?
 - b. What data or factors did you consider when making your decision?
 - c. What information were you given during that scenario?
 - d. What did you know about the situation before you were presented with this decision scenario?
 - e. What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - f. Did you decide in a team or on your own?
 - g. Once you made a decision, can you run me through how the resources were allocated?

- i. *if not covered* Who or what was responsible for the distribution of resources?
- 9. *If time permits* Have you encountered the same type of resource decision *add details based on their description* in other disaster events?
 - a. Can you tell me more about the similarities between those situations?
 - b. Can you tell me more about the differences between those situations?
- 10. *If time permits* Have you encountered different situations with a different type of limited resources and high demand? Can you please describe them for me?

Protective Actions Decision-making Scenarios

- 11. Have you participated in making decisions about protective measures (protective actions) before a forecasted disaster?
 - a. What type of protective measures were involved?
 - b. What population was affected?
 - c. In that scenario, can you please tell me about the decision process?
 - d. What data or factors did you consider when making your decision?
 - e. What information were you given during that scenario?
 - f. What did you know about the situation before you were presented with this decision scenario?
 - g. What did you know about the stakeholders (people who are affected by the protective measures) before you were presented with this decision scenario?
 - h. Did you decide in a team or on your own?
 - i. Once you made a decision, can you walk me through how the protective measures were executed?
 - i. *if not covered* Who or what was responsible for executing the protective measures?
- 12. *If not covered in previous question* Have you participated in making decisions about evacuation plans before a forecasted disaster?
 - a. What type of evacuation plans were involved?
 - b. What population was involved?
 - c. In that scenario, can you please tell me about the decision process?
 - d. What data or factors did you consider when making your decision?
 - e. What information were you given during that scenario?
 - f. What did you know about the situation before you were presented with this decision scenario?
 - g. What did you know about the stakeholders (people who are affected by the protective measures) before you were presented with this decision scenario?
 - h. Did you make a decision in a team or on your own?
 - i. Once you made a decision, can you walk me through how the evacuation measures were executed?
 - i. *if not covered* Who or what was responsible for executing the evacuation measures?
- 13. *If not covered in previous question* Have you participated in making decisions about the pre-positioning of resources before a forecasted disaster?
 - a. In that scenario, can you please tell me about your decision process as to how to decide where to pre-position resources?
 - b. *If not covered* Did you make use of logistic staging areas?
 - i. Can you tell me more about logistic staging areas?
 - c. What data or factors did you consider when making your decision?
 - d. What information were you given during that scenario?
 - e. What did you know about the situation before you were presented with this decision scenario?
 - f. What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - g. Did you decide in a team or on your own?
 - h. Once you made a decision, can you run me through how the resources were pre-positioned?
 - i. Who or what was responsible for the pre-positioning of resources?

Opportunities to Improve Decision-making Scenarios

Thank you so much for your detailed responses! Now that we've discussed some interesting decision scenarios, I'm curious about your perspective on how things could be done better. Please answer the following questions keeping in mind the decision scenarios we talked about before.

14. In the scenarios we discussed before, how effective was the amount of information/general setup of the process for you in making your decision? Why or why not?
15. In any of the scenarios we discussed before, what do you think could help improve the decision making process?
16. Now, let's focus on technology—where do you think AI technology could help improve your decision making process?
17. *If mentioned AI in survey* How would you improve the AI you already use in your job?

Disaster Recovery Center Planning

Now, we're going to shift our focus to disaster recovery. We're interested in physical locations set up after a natural disaster where survivors can learn about financial assistance programs and get help applying for assistance.

18. What term do you use to describe such locations?

If they ask for clarification about what we mean, mention these may be called Disaster Recovery Centers or Multi-agency resource centers (MARC)s

19. Have you been involved in planning or supporting the placement of these “disaster recovery centers”¹?
20. *if yes* How long after a typical natural disaster are you/your department involved in administering “disaster recovery centers”?
21. Have you been directly involved in planning/placement of “disaster recovery centers”?
22. What plans are there for placement of “disaster recovery centers” before a disaster?
23. What factors do you/your team consider when deciding locations of these “disaster recovery centers”?
24. To staff and/or maintain these “disaster recovery centers”, are there partnerships worked out with local organizations and businesses worked out beforehand? How about with larger nonprofits?
25. After a disaster, how is information about “disaster recovery centers” shared with survivors?
26. *If not covered in previous question* Does this information sharing include providing information to survivors in person?

Conclusion

Before we wrap up, do you have any final thoughts about what we discussed today that you would like to share or address?

Do you have any questions for me?

Thank you so much for your time.

Checklist for Information needed about Decision Scenarios

We designed this checklist to keep track of the important information participants tell us for each decision scenario.

If a participant does not cover some of the information we need about that scenario, we can ask followup questions. If a participant already covers the information we needed, we can skip redundant questions in the rest of the interview.

For each decision scenario they describe, we would like to document the following information:

- decision process used during the scenario
- data or factors they considered when making the decision
- information they were given during that scenario
- their prior knowledge about the situation before the scenario
- their prior knowledge about the stakeholders (people who will receive the resource) before the scenario
- decision made in a team or independently
- How decision was executed
- Who was responsible for executing decision
- When making your decision, did you consider how the plan would be executed? Was this important in making your decision?
- Population affected

¹From now on, we use “disaster recovery centers” as a placeholder for the term they identified in their response to #18